

## ***Work-Related Exhaustion and Telomere Length: A Population-Based Study***

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**Background** Psychological stress is suggested to accelerate the rate of biological aging. We investigated whether work-related exhaustion, an indicator of prolonged work stress, is associated with accelerated biological aging, as indicated by shorter leukocyte telomeres, that is, the DNA-protein complexes that cap chromosomal ends in cells.

**Methods** We used data from a representative sample of the Finnish working-age population, the Health 2000 Study. Our sample consisted of 2911 men and women aged 30–64. Work-related exhaustion was assessed using the Maslach Burnout Inventory - General Survey. We determined relative leukocyte telomere length using a quantitative real-time polymerase chain reaction (PCR) -based method.

**Results** After adjustment for age and sex, individuals with severe exhaustion had leukocyte telomeres on average 0.043 relative units shorter (standard error of the mean 0.016) than those with no exhaustion ( $p = 0.009$ ). The association between exhaustion and relative telomere length remained significant after additional adjustment for marital and socioeconomic status, smoking, body mass index, and morbidities (adjusted difference 0.044 relative units, standard error of the mean 0.017,  $p = 0.008$ ).

**Conclusions** These data suggest that work-related exhaustion is related to the acceleration of the rate of biological aging. This hypothesis awaits confirmation in a prospective study measuring changes in relative telomere length over time.